Utilization of Electricity 1: Electrical Hazards and Safety Practices

---

Introduction: The Power That Runs Things and the Danger It Hides

Electricity powers modern life. It lights homes, runs appliances, charges devices, and enables communication. Yet the same force that brings convenience also brings risk. Each year, electrical accidents cause injuries, fires, and fatalities — many of which are entirely preventable through awareness and proper safety practices.

Electrical hazards are conditions that increase the risk of electric shock, fire, or injury due to unsafe wiring, damaged equipment, or improper handling of electrical devices. Common hazards include overloading outlets, damaged insulation, wet electrical conditions, faulty wiring, and improper use of extension cords. Recognizing these hazards is the first step toward prevention.

Electrical safety practices are the behaviors, protocols, and precautions that minimize the risk of electrical accidents. These include using insulated tools, avoiding overloading circuits, keeping electrical devices away from water, and knowing emergency procedures such as turning off the power source before assisting a shock victim.

This reading material examines common electrical hazards, their causes and consequences, and the safety practices that prevent accidents. It also explores emergency responses, the importance of the Philippine Electrical Code (PEC), and the role of individuals in promoting electrical safety at home, in school, and in the community.

Part 1: Understanding Electrical Hazards

Definition of Electrical Hazard

An **electrical hazard** is any condition or situation that increases the likelihood of electric shock, fire, or injury involving electrical equipment or wiring. Hazards can arise from improper installation, damaged components, unsafe user behavior, or environmental conditions such as moisture.

Common Types of Electrical Hazards

Hazard	Description	Potential Consequences
Overloading sockets	Plugging too many devices into a single outlet or extension cord	Overheating, fire, damage to appliances
Damaged insulation; exposed live wire	Frayed, cracked, or exposed wires on cords or cables	Electric shock, short circuits, fire
Wet electrical conditions	Electrical devices or outlets near water or damp areas	Electric shock, electrocution
Faulty wiring	Improperly installed or deteriorating wiring within walls or appliances	Fire, electric shock, equipment failure
Improper use of extension cords	Using indoor cords outdoors, running cords under rugs, or daisy-chaining multiple cords	Overheating, fire, tripping hazards
Unsafe handling of electrical devices	Pulling plugs by the cord, using damaged chargers, or repairing live circuits	Electric shock, equipment damage

Warning Signs of Electrical Hazards

Warning Sign	Possible Hazard
Outlet feels warm to the touch	Overloading or faulty wiring
Flickering lights	Loose connections or overloaded circuits
Burning smell from an outlet or appliance	Overheating wires or insulation damage
Buzzing or sizzling sounds	Arcing electricity or loose connections
Sparks when plugging in a device	Short circuit or damaged plug
Frequently tripping circuit breakers	Overloaded circuit or faulty wiring

---

Part 2: Electrical Safety Practices

General Safety Practices

Practice	Why It Is Important
Avoid overloading outlets	Prevents overheating and fire; each outlet has a maximum current capacity
Use only one high-power device per outlet	High-power devices (air conditioners, heaters, microwaves) draw significant current
Unplug unused devices	Reduces risk of overheating and saves energy
Keep electrical devices away from water	Water conducts electricity, increasing shock risk

Practice	Why It Is Important
Inspect cords and plugs regularly	Damaged insulation can expose live wires
Replace damaged cords immediately	Do not use electrical tape as a permanent repair
Use extension cords temporarily only	Extension cords are not designed for permanent use
Pull plugs by the plug body, not the cord	Pulling the cord damages internal wires

Preventing Overloading

Do	Don't
Distribute appliances across multiple outlets	Plug multiple high-power devices into one extension cord
Use power strips with built-in circuit breakers	Daisy-chain multiple extension cords together
Know the power rating of appliances	Ignore warm outlets or flickering lights
Unplug devices when not in use	Use indoor extension cords outdoors

Electrical Safety in Wet Conditions

Safe Practice	Explanation
Keep electrical appliances away from sinks, bathtubs, and pools	Water significantly increases conductivity and shock risk
Use ground fault circuit interrupters (GFCIs) in wet areas	GFCIs shut off power within milliseconds of detecting a ground fault

Safe Practice	Explanation
Do not use electrical devices with wet hands	Moisture on skin reduces electrical resistance
Ensure outlets near water sources have weatherproof covers	Prevents moisture from entering the outlet

Handling Damaged Equipment

Situation	Correct Action
Frayed or cracked cord	Stop using immediately; replace the cord or the entire device
Exposed wires	Do not touch; unplug if safe to do so; have a qualified electrician repair
Overheating appliance	Unplug and have it inspected; do not use until repaired
Sparking outlet	Do not use the outlet; turn off the circuit breaker; call an electrician

---

Part 3: Emergency Response to Electrical Incidents

Responding to Electric Shock

Step	Action
1. Do NOT touch the victim directly	The victim may still be in contact with live current, making the rescuer a second victim

Step	Action
2. Turn off the power source	If possible, unplug the device or turn off the circuit breaker
3. Remove the victim from the source	Use a non-conductive object (wooden broom handle, dry rope, rubber mat) to move the victim away
4. Check for responsiveness	If unconscious or not breathing, call for emergency medical assistance immediately
5. Perform CPR if trained	Do not attempt CPR unless properly trained
6. Seek medical attention	Even mild shocks can cause internal injuries; all shock victims should be evaluated by a medical professional

Common Mistakes in Emergency Response

Mistake	Why It Is Dangerous
Grabbing the victim with bare hands	The rescuer becomes part of the electrical circuit and can be electrocuted
Using metal objects to move the victim	Metal conducts electricity
Pouring water on an electrical fire	Water conducts electricity; use a dry chemical fire extinguisher instead
Attempting to cut live wires	Extremely dangerous; only qualified electricians should handle live wires

Responding to Electrical Fires

Do	Don't
Use a Class C (or ABC) fire extinguisher	Use water
Turn off the main power if safe to do so	Attempt to unplug burning equipment
Evacuate the area if the fire spreads	Re-enter the building until declared safe
Call emergency services	Assume the fire will go out on its own

---

Part 4: The Philippine Electrical Code (PEC)

What Is the Philippine Electrical Code?

The **Philippine Electrical Code (PEC)** is a set of regulations and standards governing the design, installation, operation, and maintenance of electrical systems in the Philippines. It is designed to ensure electrical safety, prevent fires, and protect individuals from electrical hazards.

Key Provisions Relevant to Safety

Provision	Requirement
Proper grounding	All electrical systems must be grounded to prevent shock
Overcurrent protection	Circuit breakers and fuses must be appropriately sized for the circuit
Wiring methods	Wires must be properly insulated and installed according to code

Provision	Requirement
Outlet spacing	Outlets must be placed at intervals to prevent overuse of extension cords
GFCI requirements	GFCIs are required in wet locations (bathrooms, kitchens, outdoor areas)
Clearance around electrical panels	Panels must have unobstructed access for emergency shutdown

Common PEC Violations

Violation	Risk
Overloaded circuits	Overheating, fire
Unprotected wiring	Shock, short circuits
Improper grounding	Shock hazard
Use of non-certified electrical products	Substandard components may fail or overheat
Lack of GFCIs in wet areas	Increased shock risk

Part 5: Conducting an Electrical Safety Inspection

Inspection Checklist

Area	Items to Check
Outlets and Wires	Overloaded sockets, damaged/exposed wires, flickering lights
Appliances and Devices	Overheating appliances, tangled or unsafe extension cords, unplugged unused devices
Environmental Risks	Electrical items near water/damp areas, wires and devices placed unsafely, improper ventilation for electrical equipment
Emergency Readiness	Working fire extinguisher nearby, emergency contact availability, posted electrical safety procedures

Risk Level Assessment

Risk Level	Description	Action Required
Low Risk	Minimal danger; can be easily managed with basic precautions	Monitor and address during routine maintenance
Medium Risk	Moderate danger; could lead to injury or equipment damage if left unchecked	Schedule repair or corrective action within a reasonable timeframe
High Risk	Severe danger; poses immediate risk of electrocution, fire, or equipment failure	Take immediate action; restrict access to the area if necessary

Part 6: Electrical Safety in Different Settings

At Home

Hazard	Safety Practice
Overloaded outlets	Use power strips with circuit breakers; unplug unused devices
Damaged cords	Replace frayed cords immediately; do not use electrical tape as permanent repair
Children near outlets	Use outlet covers; teach children not to insert objects into outlets
Water near electrical devices	Keep appliances away from sinks and tubs; install GFCIs in bathrooms and kitchens

At School

Hazard	Safety Practice
Overloaded circuits in classrooms	Distribute appliance loads across multiple outlets; report warm outlets
Damaged extension cords	Inspect cords regularly; replace damaged ones immediately
Unsupervised use of electrical equipment	Ensure proper training before allowing students to use electrical devices
Lack of emergency procedures	Post electrical safety procedures; ensure fire extinguishers are accessible

In the Community

Hazard	Safety Practice
Illegal electrical connections (jumpers)	Report to local electric cooperative; these connections cause fires and overloads
Damaged utility poles or wires	Report to the electric company immediately; stay at a safe distance
Improper use of generators	Use only outdoors; never connect directly to household wiring without a transfer switch
Lack of public awareness	Participate in community electrical safety seminars

---

Part 7: Personal Protective Equipment (PPE) for Electrical Work

Equipment	Purpose
Insulated gloves	Protect hands from electric shock
Safety goggles	Protect eyes from sparks or arc flash
Rubber-soled shoes	Reduce risk of current passing through the body
Hard hat	Protect head from falling objects (construction sites)
Insulated tools	Prevent current from passing through the tool to the user

Important note: Only qualified electricians should perform electrical repairs or installations. Individuals without proper training should not attempt to fix live circuits or damaged wiring.

Key Terms

Term	Definition
Electrical hazard	A condition that increases the risk of electric shock, fire, or injury involving electrical equipment
Overloading	Plugging too many devices into a single outlet or circuit, causing overheating
Damaged insulation	Frayed, cracked, or exposed wire covering that can lead to shock or short circuits
Faulty wiring	Improperly installed or deteriorating electrical wiring
Electrocution	Death caused by electric shock
Ground Fault Circuit Interrupter (GFCI)	A device that shuts off power within milliseconds when it detects a ground fault
Short circuit	An unintended path of low resistance that allows excessive current to flow
Philippine Electrical Code (PEC)	The set of regulations governing electrical system safety in the Philippines
Personal Protective Equipment (PPE)	Safety gear worn to protect against electrical hazards

Review Questions

Responses should be written on a separate sheet of paper.

1. List five common electrical hazards and describe one safety practice to prevent each.
2. A classmate receives a mild electric shock after touching a damaged outlet. What is the first action that should be taken, and why?
3. Explain why extension cords should not be used as permanent wiring. What are the risks of daisy-chaining multiple extension cords?
4. What is the purpose of the **Philippine Electrical Code (PEC)**? Give three examples of requirements it mandates.
5. Describe how you would conduct a basic electrical safety inspection in your classroom. What would you look for, and what actions would you take if you found a hazard?